

Jacobs Industries

Multi-Region Supply Chain Expansion & Wind-Down Plan - Business Report

1. Executive Summary

Jacobs Industries manufactures an industrial insulating foam chemical. Today the product is sold in only one region, Calopeia, through a single factory and a single warehouse, with every customer order fulfilled by mail. Four additional regional markets - Sorange, Tyran, Fardo, and Entworpe - have known, measurable demand but are not yet served by any dedicated local infrastructure. A competing technology will make this entire product line obsolete on a fixed date (day 1,460 of the planning horizon, with demand decaying to zero over the final 30 days). The business objective is to maximize cash generated before that date.

This report answers three linked questions: which of the four new markets Jacobs should serve and when, how much (if any) new factory and warehouse capacity to build in each region, and how to run production and inventory efficiently - while making sure nothing is left unsellable when the product is discontinued.

Recommendation summary:

Serve all five regions. Starting immediately (day 730), expand the existing Calopeia factory from 70 to roughly 168 drums/day (~\$4.9M) so it can keep supplying Calopeia, Sorange, and Tyran; build new local warehouses in Sorange and Tyran (ready by day 790); and build both a new factory and a new warehouse in Fardo (ready by day 820), since Fardo's freight costs are high enough that local production and stocking pay for themselves. Entworpe is the one market that is best left unserved by dedicated local infrastructure - its demand arrives as random, fixed-size 250-drum bursts that a stocked local warehouse cannot serve efficiently, so it stays on mail fulfillment directly from the Calopeia warehouse, backed by a standing 250-drum buffer. Across the network, factory-to-warehouse shipments always move by truck (cheaper per drum than mail at full loads) and

all last-mile customer deliveries move by mail (the only available channel). Production batch sizes and reorder points are tuned per route using demand-and-cost based (newsvendor) logic rather than a flat target, and the whole network is deliberately wound down in the final 30 days - reorder points cut by half on day 1,430, all production stopped by day 1,445 - so that Jacobs is not left holding inventory that will never sell.

2. Market Entry Decisions

All five regions already generate demand from day 730 onward. The real decision is not whether to accept orders from each region (mail fulfillment from the Calopeia warehouse can technically reach all five from day one), but where to invest in dedicated local factory and/or warehouse infrastructure so that Jacobs can serve each region's demand profitably and reliably through to the end-of-life date.

Table 1

Decision by region

Region	Continue / Enter?	Local infrastructure	Ready by
Calopeia	Continue (home market)	Existing factory, expanded to ~168 drums/day	Day 820
Sorange	Enter	New warehouse (supplied by Calopeia factory via truck)	Day 790
Tyran	Enter	New warehouse (supplied by Calopeia factory via truck)	Day 790
Fardo	Enter	New factory + new warehouse	Day 820
Entworpe	Enter (mail-only)	None - served from Calopeia warehouse, 250-drum standing buffer	Day 730

Forecast daily demand ramps up over the two-year historical/forecast cycle before settling into a repeating pattern. The combined five-region total ranges from about 84 drums/day in the trough phase to about 213 drums/day at peak (workbook “Summary” sheet, Total/day column), and the network is expected to move roughly 111,030 drums in total over the full day 731-1460 horizon.

Table 2

Forecast demand by region - remaining life of the product (day 821-1429)

Region	Forecast volume (drums)
Calopeia	25,760
Sorange	45,241
Tyran	11,821
Fardo	9,564
Entworpe	7,613

3. Demand Forecasting Approach

Each region’s demand behaves differently, so a different forecasting technique was used for each - matched to the actual shape of that region’s demand history rather than applying one method everywhere.

Calopeia - seasonal index

Calopeia’s two years of historical data show a repeating ~365-day cycle (trough → ramp-up → peak → decline → trough) with no long-run growth trend, so a seasonal-index method was used: each of phase was assigned an index relative to the overall average, and that index was applied to the most recent year’s average level to produce the forecast.

Table 3

Calopeia - seasonal index

Phase	Seasonal index	Forecast (drums/day)
1st Trough	0.43	16.5
Ramp up	1.19	46.1
Peak	1.76	67.9
Decline	1.25	48.3
2nd Trough	0.37	14.3

Sorange - linear regression

Sorange's demand grows steadily over time with no seasonality, so an OLS linear regression trendline was fit to day 640-730 data (slope = 0.153, intercept = 0.099), then extended forward. The forecast climbs from about 20 drums/day (1st Trough, year 1) to 118 drums/day by the 2nd Trough of year 2, before decaying to zero by day 1,460.

Tyran and Fardo - historical average (excluding ramp-up)

Both regions' demand started near zero around day 640 and ramped up over about 30 days to a stable level, with no further trend or seasonality after that. A simple average that included the ramp-up days would underestimate the true steady-state level, so only day 670-730 (post-ramp-up) was averaged. This produces a flat forecast of 19 drums/day for Tyran and 16 drums/day for Fardo for every phase, each decaying to roughly half that level (10 and 8 drums/day) during the day 1,430-1,460 wind-down.

Entworpe - inter-arrival time model

Entworpe does not order continuously; it places fixed-size 250-drum orders at random intervals (a (Q, R) reorder policy on the customer's side). Over the 60-day clean history window (day 670-730), 3 orders were observed, for an average inter-arrival time of 20 days. Dividing the fixed order size by the inter-arrival time gives

an underlying consumption rate of $250 \div 20 = 12.5$ drums/day (1.5 orders per month), used as a flat forecast, decaying to 6.25 drums/day (0.75 orders/month) during wind-down. Because the order timing is random and cannot be forecast more precisely, the operating plan is not to predict exactly when each order will land, but to keep a standing 250-drum buffer in the Calopeia warehouse at all times so any burst order can be filled immediately.

4. Factory Capacity Decision

The existing Calopeia factory has a capacity of 70 drums/day. Expanding capacity takes a 90-day construction lead time, so the decision needed to be made immediately (day 730) rather than waiting to confirm demand.

Account for 90-day build period

The model explicitly quantifies the cost of the 90-day build delay: at the original 70 drums/day capacity, production over that window would total 6,300 drums against forecast demand of 8,233 drums - a shortfall of about 1,933 drums, which is covered by the starting inventory of 2,528 drums at day 731.

Sizing the expansion

After day 820 (when the 90-day build completes), the factory has 610 productive days remaining before wind-down begins on day 1,430. Given the plan to build Fardo its own dedicated factory (Section 5), Calopeia's factory only needs to keep covering Calopeia, Sorange, Tyran and Entworpe demand. The minimum capacity needed for those three routes is 153 drums/day; adding a 10% safety buffer (to protect against forecast error without needing to "prebuild" inventory during troughs) sets the target capacity at 168 drums/day.

Decision: expand the existing Calopeia factory from 70 to 168 drums/day (an increase of about 98 drums/day, roughly +140%), at a cost of \$50,000 per unit of daily capacity - approximately \$4.91M ($98 \times$

\$50,000). Expanding the existing factory was preferred over building an entirely separate one for this demand because it avoids the \$500,000 fixed cost of a brand-new factory.

5. New Factory Decisions

For each new region, the model compares building a dedicated local factory (plus the warehouse and capacity needed to run it) against continuing to ship product in from the expanded Calopeia factory. The per-drum shipping cost saved by producing locally is weighed against the fixed cost of a new factory (\$500,000), the variable cost of building enough local capacity (\$50,000/unit of daily capacity), and a new warehouse (\$100,000).

Table 4

New factory decisions

Region	Saving/drum	Local capacity needed	Total build cost	Breakeven volume	Forecast volume	Net gain	Decision
Sorange	\$50	74/day	\$4,308,292	86,166	45,241	(\$2,046,234)	Skip
Tyran	\$50	19/day	\$1,568,901	31,378	11,821	(\$977,871)	Skip
Entworpe	\$50	12/day	\$1,223,975	24,480	7,613	(\$843,350)	Skip
Fardo	\$250	16/day	\$1,383,959	5,536	9,564	+\$1,007,115	Further check

For Sorange, Tyran, and Entworpe, the per-drum saving from local production is only \$50 (these regions are on the same continent as Calopeia, so freight is not that expensive to begin with), and none of the three

generate enough remaining volume to earn back a ~\$1.2-4.3M investment before the product's end-of-life - all three show a negative net gain, so a new factory is skipped in each.

Fardo is different: it sits far enough away that freight is more expensive per drum (same-region mail is \$150/drum vs. \$400/drum to Fardo; same-region truck is \$15,000/200-drum truckload vs. \$45,000 to Fardo), so local production saves \$250/drum instead of \$50. That saving, combined with Fardo's relatively small required capacity (16 drums/day), produces a positive net gain of about +\$1.01M - marked "Further check" pending comparison against a warehouse-only alternative (Section 6).

6. New Warehouse Decisions

A second comparison tests a lighter-weight option: building only a local warehouse (still supplied by truck from the Calopeia factory) rather than a full local factory. This captures the saving from switching the last-mile leg from long-distance mail to local mail, without the cost of local production capacity.

Table 5

New warehouse decisions

Region	Saving/drum	Breakeven volume	Forecast volume	Net gain	Decision
Sorange	\$25	4,000	46,007	+\$1,050,166	Warehouse only
Tyran	\$25	4,000	12,403	+\$210,072	Warehouse only
Entworpe	\$25	4,000	7,988	+\$99,688	Skip
Fardo	\$100	1,000	10,035	+\$903,544	Factory + warehouse beats

Sorange and Tyran both clear their breakeven volume comfortably (46,007 and 12,403 drums forecast vs. a 4,000-drum breakeven each), producing net gains of roughly +\$1.05M and +\$210K - build a warehouse in each, supplied by truck from the expanded Calopeia factory.

Entworpe technically clears breakeven too (+\$99,688 net gain on paper), but the decision is to skip it. The reason is qualitative and is documented directly in the model: Entworpe's demand arrives as random, fixed-size 250-drum bursts, which cannot be pre-positioned efficiently against a smooth reorder-point policy - in practice this creates a real risk of idle inventory sitting in a dedicated warehouse between infrequent orders, thinning out the on-paper margin. Entworpe is instead served from the Calopeia warehouse's standing 250-drum buffer (Section 3).

For Fardo, this warehouse-only option (+\$903,544 net gain) is weaker than pairing a warehouse with a local factory (+\$1,007,115 net gain from Section 5), because local production also halves Fardo's replenishment lead time (from 14 days of cross-continent shipping to about 7 days) and lowers the safety stock needed to protect against stockouts. Final decision: build both a factory and a warehouse in Fardo.

7. Production Scheduling and Inventory Policy

Once the network footprint above (Section 2) is set, each route needs its own production batch size and reorder point (ROP), sized around three inputs: revenue of \$1,450/drum, variable production cost of \$1,000/drum, a \$1,500 setup cost per production run (regardless of size), a holding cost of \$100/drum/year, and a 10% annual opportunity cost of cash tied up in inventory.

A live template, not a one-time calculation

The batch-size / safety-stock / ROP table below is a live decision tool, not a one-time snapshot: if a phase's actual demand turns out to differ from the forecast, updating that phase's figure in the "Consolidated data" sheet automatically recalculates the EOQ, practical batch, safety stock, and ROP for every affected route.

The figures below are the day-820-onward baseline calculated from the original forecast; the same template should be re-run with updated demand inputs by phase, or at any point during the game, so ordering decisions stay current rather than locked to the initial plan.

Table 6

Final batch sizes and reorder points (effective day 820 onward)

Route	Practical batch	Cycle length (days)	Target service level	Safety stock	ROP
Calopeia (+ Entworpe)	200 drums	4	99.0%	470	1,171
Sorange	400 drums	5	97.9%	67	1,029
Tyran	400 drums	21	92.5%	87	339
Fardo	400 drums	25	93.8%	148	658

Why target service levels are high (92-99%), not a flat assumption

Rather than assuming an arbitrary fixed service level, each route's target service level was set using a newsvendor / critical-ratio calculation: target SL = $C_u \div (C_u + C_o)$, where C_u is the lost margin from a stockout (\$146-\$221/drum across routes) and C_o is the cost of holding one cycle's worth of extra inventory (\$2.11-\$14.51/drum per cycle). Because C_u is so much larger than C_o on every route, the critical ratio comes out high across the board - it is far more costly to run out of stock and lose a \$146-\$221 sale than to carry a little extra inventory that costs a few dollars per cycle to hold.

Why Calopeia carries the most safety stock relative to demand

Calopeia has the second highest margin per drum (\$217.50) and the lowest holding cost/day (\$0.57), which pushes its critical-ratio service level to 99.0% - the highest of any route - and drives its safety stock of

470.0 drums, the largest in the network in absolute terms. This is deliberate: Calopeia's route also backstops Entworpe's unpredictable 250-drum orders, so it is run with a smaller, faster-turning practical batch (200 drums - one full truckload, actually below its raw EOQ of 534 drums) precisely so the production line is never tied up too long to react quickly to an Entworpe spike.

Why shared-factory routes need extra buffer time

Calopeia (+Entworpe), Sorange, and Tyran are all produced on the same expanded Calopeia production line. The model explicitly accounts for this: when one route's batch is queued behind the other two, its wait time is not just its own production time but the sum of all three routes' production times. That is why all three routes share the same adjusted production lead time (6 days) and effective lead time (13 days) - each has to budget for its neighbors' production runs finishing first, not just its own. Batch sizes were deliberately kept close to one or two truckloads (200-400 drums) rather than pushed up to the larger raw EOQ values (534-622 drums) specifically to avoid one route's oversized batch blocking the other two routes on a shared line.

Fardo, by contrast, has its own dedicated (but small) production line, so it does not queue behind other routes - its longer lead time (26 days adjusted, 33 days effective) instead comes from running a low-capacity 17 drums/day line long enough to complete a full 400-drum batch, plus its own shipping time.

8. Transportation Mode

Two shipping legs exist in the network: factory → warehouse, and warehouse → customer. Every customer order must ship by mail (the only channel available for last-mile fulfillment, and it must go out within 24 hours of the order, or the sale is permanently lost). For the factory-to-warehouse leg, both truck and mail are technically available, and the decision is to always use truck, because a full 200-drum truckload is cheaper per drum than mail on every route.

Table 7

Transportation Mode

Route type	Truck cost/drum	Mail cost/drum	Truck transit	Mail transit
(200-drum load)				
Same region	\$75	\$150	7 days	1 day
Different region, same continent	\$100	\$200	7 days	1 day
Between continent and Fardo	\$225	\$400	14 days	2 days

Truck is roughly half the per-drum cost of mail on every route type, so all factory-to-warehouse shipments use truck. Mail is faster but reserved for the warehouse-to-customer leg, where it is the only permitted fulfillment channel.

9. Cash Flow Feasibility

The combined infrastructure investment implied by the recommendations above totals approximately \$6.57M > starting-cash balance of \$6.80M at day 731.

Table 8

Cash Flow Feasibility

Cash flow	Balance (\$M)
Starting	6,797,000
Expand the Calopeia	4,906,178
Build the Fardo factory + capacity + warehouse	1,462,354

Build Sorange & Tyran warehouses	200,000
Remaining working capital	228,467

10. End-of-Life Wind-Down Plan

Demand across all five regions decays linearly to zero between day 1,430 and day 1,460, the fixed date the competing technology renders the product obsolete. Any drum produced too close to the end is at risk of “stranding” - costing \$1,000/drum in variable production cost but arriving to near-zero demand and \$0 realizable value. The wind-down plan is timed to avoid this.

Table 9

End-of-life Wind-Down Plan

Day	Action
1,430	Cut all reorder points by 50% network-wide (Entworpe’s standing buffer reduced from 250 to 125 drums)
1,445	Stop all production at both factories (Calopeia and Fardo)
1,460	Product line ends

Halving reorder points on day 1,430 lets on-hand inventory run down in step with the collapsing demand curve instead of continuing to restock at full-size levels. Stopping production on day 1,445 leaves a 15-day window for the last batches produced to reach customers before day 1,460, without adding fresh production that would have almost no time left to sell. Each region’s forecast for the day 1,430-1,460 window is roughly half of its steady-state rate (e.g., Calopeia 7 vs. ~14-68 drums/day across phases; Tyran 10 vs. 19; Fardo 8 vs. 16; Entworpe 6 vs. 13), consistent with the linear decay to zero.

11. Summary Table - All Major Decisions

Decision area	Outcome	Rationale
Serve Calopeia (home market)	Continue	Existing profitable core market
Serve Sorange, Tyran, Entworpe (new)	Enter, via mail from Calopeia WH from day 730	Demand exists immediately; dedicated local build evaluated separately
Serve Fardo (new)	Enter, with dedicated local factory + warehouse by day 820	High freight cost to Fardo makes local build pay for itself
Expand Calopeia factory	Yes: 70 → 168 drums/day (~\$4.91M)	Covers Calopeia + Sorange + Tyran demand with 10% buffer; avoids \$500k new-factory cost; 90-day lead time means decision can't wait
Build new factory - Sorange	No (Skip)	Net loss -\$2.05M; not enough remaining volume to repay the investment
Build new factory - Tyran	No (Skip)	Net loss -\$0.98M
Build new factory - Entworpe	No (Skip)	Net loss -\$0.84M
Build new factory - Fardo	Yes	Net gain +\$1.01M; freight saving of \$250/drum vs. \$50 elsewhere
Build new warehouse - Sorange	Yes	Net gain +\$1.05M

Build new warehouse - Tyran	Yes	Net gain +\$0.21M
Build new warehouse - Entworpe	No (Skip)	Random 250-drum burst orders make a stocked local warehouse impractical despite a small positive net gain on paper
Build new warehouse - Fardo	Yes, paired with local factory	Factory + warehouse (+\$1.01M) beats warehouse-only (+\$0.90M); also halves replenishment lead time
Transportation - factory to warehouse	Truck, every route	Truck is ~half the per-drum cost of mail at full 200-drum loads on every route
Transportation - warehouse to customer	Mail, every route	Only channel available for last-mile fulfillment (fixed rule)
End-of-life wind-down	Cut ROPs 50% on day 1,430; stop production day 1,445	Avoids stranding high-cost inventory as demand collapses to zero by day 1,460

Prepared from analysis/supply-chain-model.xlsx (Summary, Consolidated data, and per-region sheets).

All figures above are pulled directly from the workbook's cached calculated values unless explicitly marked as a derived calculation.